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Adaptive network anomaly detection using machine learning approaches

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Abstract

This research aims to develop a Network Detection System (NDS) utilizing various machine learning techniques to enhance network security through anomaly detection. It evaluates the effectiveness of K-nearest neighbors (KNN), gradient boosting, support vector machines (SVM), random forests, and logistic regression in identifying deviations from normal network behavior. Furthermore, ensemble learning methods, including voting and stacking techniques, are explored to improve detection accuracy. The study proposes and tests a hybrid multi-layered stacking model using the CICIDS 2017 dataset, which encompasses both historical and modern attack patterns, providing a comprehensive benchmark for evaluation. Model performance is assessed using metrics such as accuracy, precision, recall, and F1 score. Special emphasis is placed on feature importance and reduction in dimensionality to enhance model efficiency. Additionally, the study addresses the critical challenge of minimizing false positives and false negatives for practical deployment. Results indicate that the hybrid ensemble stacking model achieves superior performance, with an accuracy of 98.79%, significantly improving network anomaly detection. The research highlights the potential for further advances through deep learning and real-time detection methodologies to improve network security in the future.

Keywords Network detection system (NDS), Ensemble learning, Voting, Anomaly detection, CICIDS2017

1 Introduction

In the modern digital era, where network security is paramount, the surge in cyberattacks has highlighted the need for more advanced anomaly detection methods. Conventional network security solutions, which often depend on static rule-based systems, frequently fall short

in identifying complex and evolving threats [1]. Machine learning has emerged as a powerful alternative, offering the capability to learn from data and adapt to new attack patterns autonomously, without requiring continuous human oversight [2]. This field has seen rapid growth, with numerous studies demonstrating the effectiveness of machine learning algorithms in detecting network threats while reducing false positives [3].

This research focuses on detecting a range of network attacks, including Distributed Denial of Service (DDoS), port scanning, and various denial of service (DoS) attacks such as DoS Hulk, GoldenEye, Slowhttptest, and Slowloris. Each type of attack presents distinct challenges: DDoS floods systems with excessive traffic, disrupting service availability, while Port Scanning systematically probes network ports to uncover vulnerabilities or open services, often serving as a precursor to more severe attacks. DoS Hulk is an HTTP-based attack that generates a massive volume of web traffic to deplete server

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resources, and GoldenEye similarly exploits HTTP vulnerabilities to exhaust server capabilities. Slowhttptest and Slowloris, categorized as “slow” DoS attacks, extend the duration of incomplete HTTP requests, monopolize server connections and deny legitimate user access. Detecting these attacks effectively is critical to preserving network integrity and enabling a prompt response to incidents.

The recent research explores machine learning algorithms applied to network security and strengthening protection through ML techniques [4]. Artificial intelligence, big data, and emerging techniques like cloud integration and transfer learning enhance smart grid cybersecurity by enabling early threat detection, proactive response, and greater resilience [5]. To improve detection accuracy and robustness, this study utilizes a multilayer stacking model, which combines the predictive strengths of multiple base classifiers with a meta-classifier. This ensemble technique, particularly stacking, has proven effective in enhancing anomaly detection systems by capitalizing on the strengths of individual algorithms [6]. By integrating multiple models through this approach, the research aims to develop a more resilient detection system capable of addressing the complex and dynamic nature of network threats. This multilayer strategy ensures greater adaptability and precision in classification, contributing to the advancement of next-generation Network Detection Systems (NDS) [7].

1.1 Scope of the proposed work:

- Advanced threat detection: enhances the capability to detect sophisticated and evolving cyber threats that traditional methods may miss, providing a more adaptive and responsive approach to network security.
- Improved accuracy: focuses on reducing false positives and false negatives, which is crucial for minimizing unnecessary alerts and ensuring that real threats are detected accurately, thereby improving operational efficiency.
- Application of machine learning: demonstrates the effectiveness of ML methods in network anomaly detection, which can handle complex patterns and dynamic network behaviour better than static rule-based systems.
- Utilization of real-world dataset: leverages dataset CICIDS 2017, ensuring that the models are evaluated against both historical and current attack patterns, thus providing more relevant and generalizable results.

- Ensemble learning techniques: explores the potential of ensemble methods, specifically voting and stacking, to improve model performance by combining the strengths of multiple algorithms, leading to better detection accuracy.

In this article, Sect. 1 introduces the paper, outlining its background, significance, and objectives, emphasizing the importance of anomaly detection in network security and the potential advantages of machine learning-based solutions. Section 2 provides a review of the literature that summarizes previous research on network anomaly detection, with a focus on existing approaches, algorithms, and techniques that highlight the challenges and limitations of traditional detection methods. In Sect. 3, the research methodology is detailed, covering model design, feature engineering, and data processing techniques. This chapter explains the chosen machine learning models, the feature selection methods, and the data preprocessing steps used to prepare the data sets. In Sect. 4, the results of the study are analyzed, with an emphasis on comparing the performance of the hybrid ensemble model to individual algorithms and demonstrating the advantages of ensemble methods in enhancing detection accuracy. Finally, Sect. 5, concludes the paper by summarizing the findings, contributions, and suggestions for future research, including potential avenues for further exploration of deep learning techniques and real-time detection capabilities.

2 Literature review

Anomaly detection research in cloud environments has grown significantly, emphasizing the need to strengthen security in complex and interconnected networks. For instance, Hagemann and Katsarou [1] surveyed machine learning techniques designed specifically for cloud networks, showcasing a wide range of applications in areas such as intrusion detection and performance monitoring. Their work underscores the importance of feature selection and scalability in ensuring model effectiveness. Expanding on this foundation, Shahzad et al. [2] explored ensemble learning for multiclass anomaly detection, employing methods like decision trees and gradient boosting to enhance accuracy and robustness in cloud-based systems. Gupta et al. [3] tackled the challenges of multi-cloud environments by introducing a federated learning approach that addresses privacy concerns while ensuring robust anomaly detection. Integrated with digital twins, their hierarchical federated learning framework facilitates collaborative model training across multiple platforms without exposing sensitive data, offering a compelling solution for cloud-based applications. In network traffic analysis, Kurniabudi et al. [6] utilized

information gain for feature selection on the CICIDS2017 dataset to enhance machine learning models, striving to achieve a balance between detection accuracy and computational efficiency. Similarly, Maseer et al. [7] conducted an extensive benchmarking of machine learning algorithms using the same dataset, emphasizing metrics such as accuracy and detection rates to determine the most effective approaches for anomaly detection.

Additional research underscores the need to balance performance with practical deployment requirements. For example, the study “Evaluating the Performance and Challenges of Machine Learning Models in Network Anomaly Detection” [8] examines various models, focusing on challenges such as feature selection, computational overhead, and adaptability to real-world scenarios. Likewise, Neuschmied et al. [9] proposed a two-stage intrusion detection framework that integrates lightweight anomaly detection with a secondary, detailed analysis, effectively reducing false positives while enhancing detection efficiency. Adnan, Bshara, and Awad [10] explored the use of unsupervised machine learning for forensic analysis of Advanced Persistent Threat (APT) attacks, emphasizing adaptability in detecting APT patterns without relying on labeled data. Their study addresses key challenges in feature extraction and real-time threat detection, which are critical for cybersecurity forensics. Similarly, Abdallah and Ootom [11] conducted a survey on supervised machine learning for intrusion detection, comparing different methodologies across various environments and providing valuable insights for future research directions.

Noto et al. [12] highlighted the effectiveness of ensemble methods in anomaly detection, showing that integrating multiple feature models improves robustness, particularly in noisy datasets. Similarly, Vanerio and Casas [13] emphasized the advantages of ensemble techniques in reducing false positives, especially in complex network environments, and stressed the need for careful model selection to optimize detection performance. Shon and Moon [14] demonstrated that hybrid approaches, which integrate multiple machine learning algorithms, offer greater consistency and reliability in anomaly detection compared to single-model methods. Lok et al. [15] proposed a hybrid model that combines several algorithms to improve flexibility and accuracy in detecting unusual data patterns. Alkasasbeh [16] introduced an innovative hybrid method that integrates traffic prediction with change point detection, designed to enhance real-time anomaly detection in dynamic network environments. Wang et al. [17] conducted a comprehensive evaluation of various algorithms and methods used in practical anomaly detection applications, addressing challenges like scalability, adaptability, and the demand

for continuous refinement. Their study provides an insightful outlook on leveraging emerging technologies to improve the effectiveness of anomaly detection systems.

The authors in [18] presents innovative strategies to improve intrusion detection systems (IDSs) by overcoming limitations in existing approaches, building on prior studies that have employed machine learning techniques such as nature-inspired metaheuristics, Bayesian methods, and swarm intelligence for attribute selection and performance enhancement. The authors proposed an XAI-enhanced ensemble model that improves transparency by explaining detection decisions, while remaining scalable and efficient for real-time deployment on resource-constrained IoT gateways [19]. Blockchain and Federated Learning (FL) provide complementary strengths that make them particularly effective for secure, privacy-preserving, and collaborative network anomaly detection. Federated learning enables multiple organizations or distributed devices to collaboratively train machine learning models without directly sharing sensitive raw data, thereby mitigating privacy and regulatory concerns. However, FL alone is susceptible to challenges such as model poisoning, data integrity issues, and a lack of trust among participating entities. Blockchain, with its decentralized ledger, immutability, and consensus mechanisms, addresses these limitations by ensuring transparent auditing, traceability, and tamper-resistant record-keeping of model updates and transactions. The synergy of Blockchain and FL thus creates a trustworthy ecosystem for collaborative security analytics. This dual approach is especially beneficial in three contexts:

- Multi-organization cybersecurity collaborations—enterprises, governments, and research institutions can jointly detect and mitigate large-scale cyber threats without exposing sensitive internal data. Blockchain ensures fairness and accountability, while FL enables knowledge sharing across organizational boundaries.
- IoT/edge networks with limited resources—in highly distributed environments such as IoT or edge computing, FL reduces the need for centralized data transfer, while Blockchain ensures secure coordination among heterogeneous devices, even in untrusted settings.
- Privacy-sensitive sectors like healthcare and finance—both domains demand stringent compliance with privacy regulations (e.g., HIPAA, GDPR). FL allows cross-institutional anomaly detection without centralizing confidential records, while Blockchain enforces secure access control and verifiable model updates.

Recent studies have highlighted the promise of this integration. For example, Buyuktanir et al. demonstrated how Federated Learning improves network intrusion detection systems [20], while Khawla et al. emphasized the role of blockchain-based intrusion detection systems in IoT networks [21]. Kang et al. demonstrated a blockchain-enabled federated learning framework to detect anomalies in mobile edge computing environments. By leveraging blockchain's immutable ledger, participants in different telecom operators could collaboratively train intrusion detection models without exposing raw traffic data. The system improved anomaly detection accuracy compared to traditional federated learning while reducing trust risks in multi-operator collaborations [22]. Danquah et al. implemented an efficient FL-based framework with reduced computational complexity for effective detection of IoT botnet threats [23]. Zhang et al. outline a typical blockchain-based decentralized federated learning framework, detailing its operational workflow and highlighting applications across domains such as the Internet of Things and the Internet of Vehicles [24].

The study in this article seeks to provide key insights into the application of machine learning and ensemble methods to enhance network security [1, 2], with a forward-looking focus on integrating deep learning and real-time detection capabilities [3, 6]. By investigating the potential of ensemble techniques and introducing a novel hybrid model, the research aims to make meaningful contributions to the field [7–9]. It emphasizes the benefits of machine learning in improving anomaly detection accuracy [10, 11] while acknowledging the importance of further exploration into deep learning and real-time

detection [12, 13]. The findings offer valuable guidance for the design of next-generation network detection systems [14, 15], contributing to the development of a more secure and robust digital infrastructure [16, 17].

A notable gap exists in the application of stacking algorithms for anomaly detection, especially using the CICIDS2017 dataset. Specifically, multi-layer stacking techniques, which are valuable for optimizing multi-class and imbalanced data, remain underexplored. This highlights the need for research into deeper stacking layers capable of capturing complex interdependencies among diverse feature sets. In addition, there is a scarcity of studies on lightweight stacking models designed to balance efficiency and accuracy, making them suitable for real-time detection in resource-constrained environments.

3 Methodology

The proposed methodology aims to investigate the research hypothesis outlined in Fig. 1. It includes the following different stages:

3.1 Data collection

In the initial step, we loaded and pre-processed the CICIDS2017 dataset, a widely recognized benchmark for anomaly detection systems. This dataset includes an extensive collection of normal and malicious network traffic records. Using the Pandas library, the data was imported into a Python environment. We also used another dataset, NFS-2023-TE which is a revised version of CICIDS2017 for experimental evaluation.

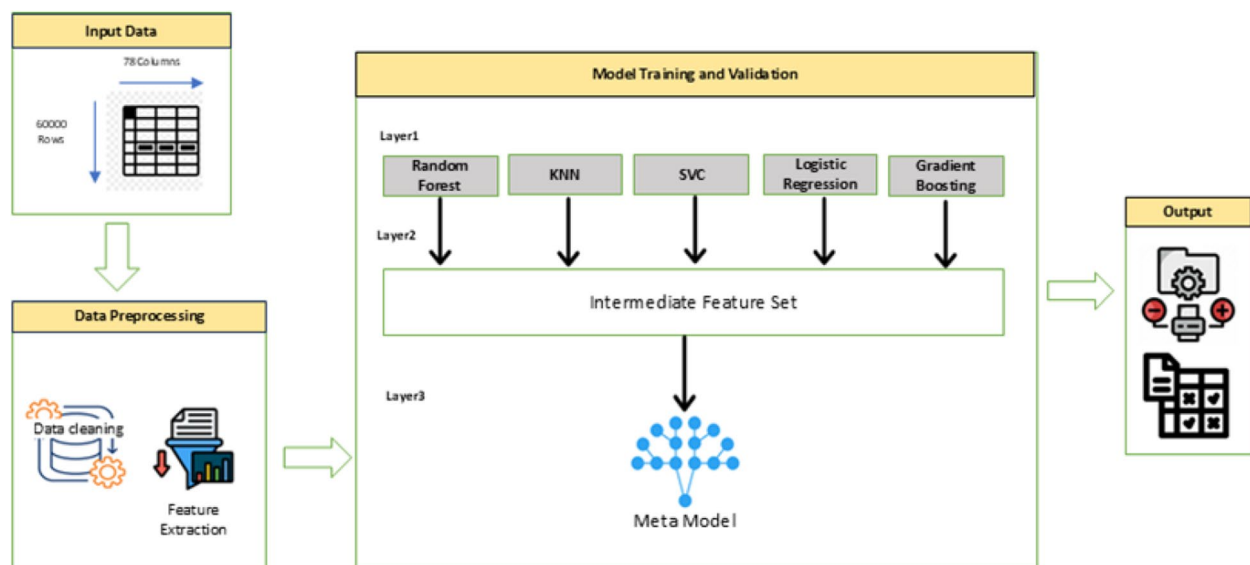


Fig. 1 Data flow diagram of proposed methodology

3.1.1 Data splitting

Given the large size of the original dataset, which comprises 2,830,744 rows, it was reduced to 60,000 rows for this analysis. The data was then divided into training, testing, and validation sets using the `train_test_split` function from scikit-learn. Specifically, 70% of the dataset was allocated for training, 20% for testing, and the remaining 10% for validation.

3.2 Data preprocess and feature importance

For tree-based models such as Random Forest, feature importance was analyzed in greater detail to identify the most critical and influential features for network anomaly detection. The feature importance attribute of the Random Forest model was employed to evaluate the relative significance of individual features within the dataset. A visual representation of feature importance was created, providing valuable insights and aiding in the prioritization of the most impactful variables for anomaly detection. Figure 2 illustrates the top features selected by the SelectKBest method for CICIDS2017 dataset, ranked according to their importance scores. The x-axis represents the importance scores, while the y-axis lists the corresponding features, with Bwd Packet Length Mean and Avg Bwd Segment Size identified as the most significant. This analysis facilitates the identification of key features that enhance the performance of machine learning

models. Figure 3 illustrates similar results for NFS-2023-TE dataset used.

3.3 Model selection

To construct an ensemble model, we selected five diverse machine learning algorithms: random forest (RF), gradient boosting (GB), K-nearest neighbors (KNN) classifier, support vector machine (SVM), and logistic regression (LR). These algorithms were chosen to provide a comprehensive combination of methods capable of capturing both linear and non-linear relationships. Furthermore, including ensemble techniques, such as voting and stacking, ensures the development of a robust predictive framework.

3.4 Ensemble learning model

To leverage the strengths of individual models, two types of ensemble classifiers were implemented using the Voting Classifier from the scikit-learn library:

3.4.1 Voting

This method was employed with both hard and soft voting strategies. The Hard Voting Classifier is particularly effective when the base classifiers exhibit similar levels of reliability, as it selects the majority class based on discrete outputs. The Soft Voting Classifier is advantageous when base classifiers provide probabilistic outputs, enabling

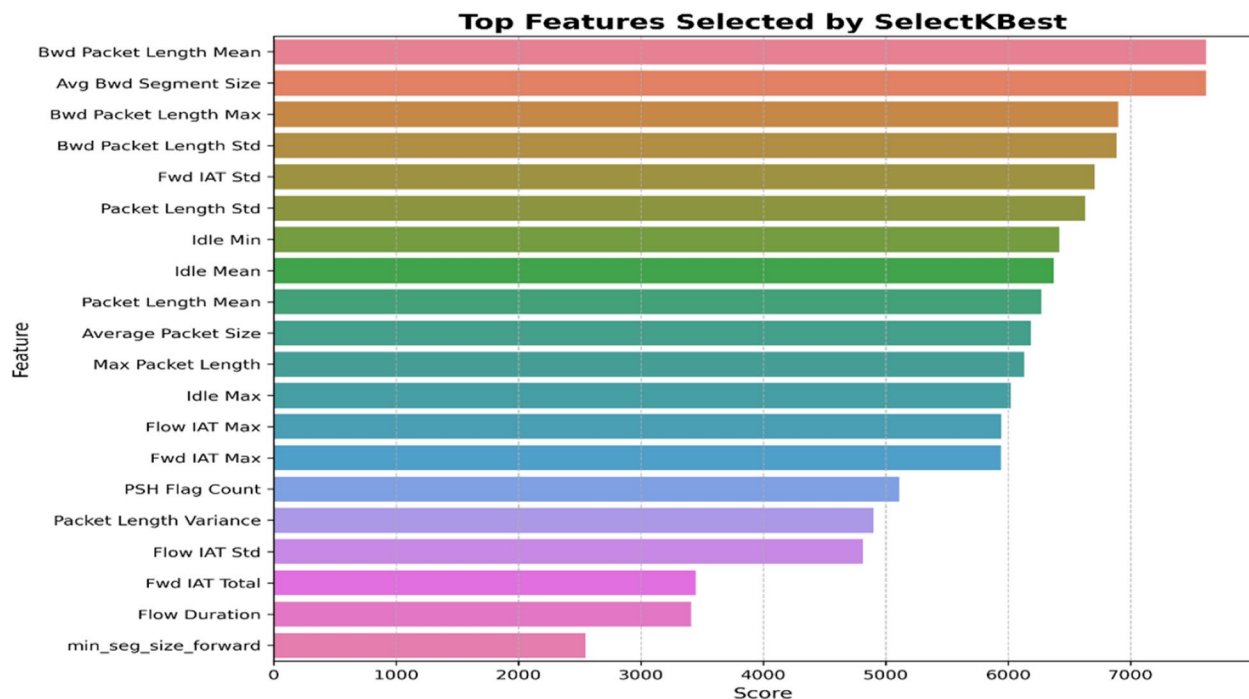


Fig. 2 Feature importance from CICIDS2017 dataset

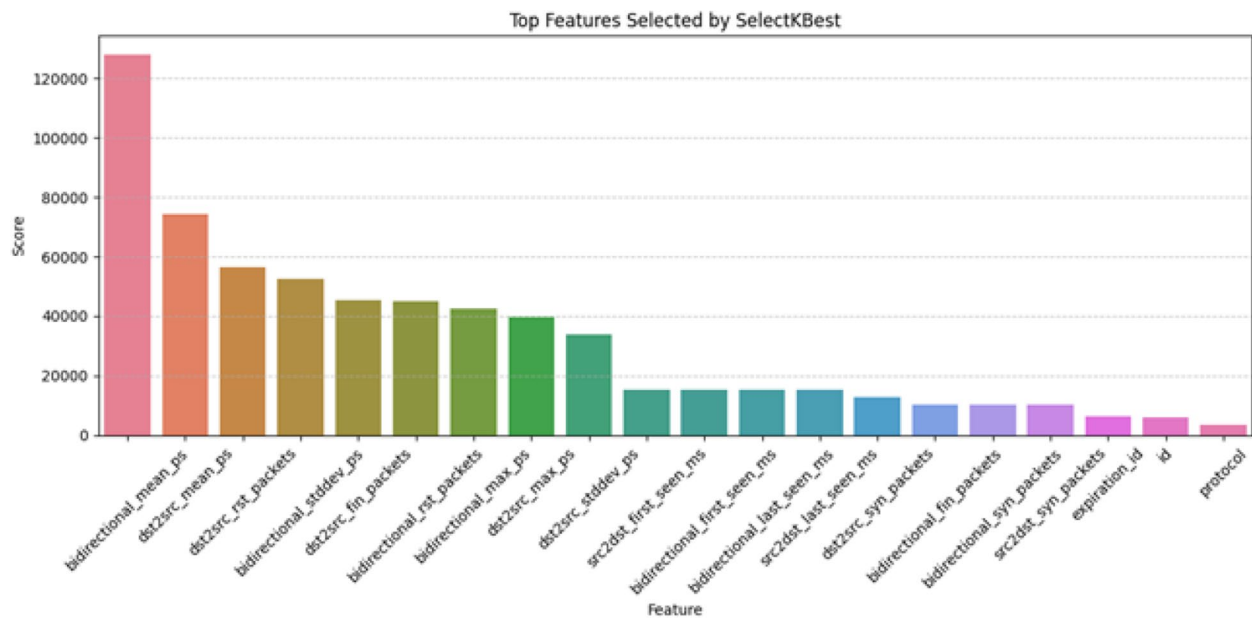


Fig. 3 Feature importance from NFS-2023-TE dataset

more informed decision-making through weighted averaging of predictions.

3.4.2 Stacking

Stacking combines the predictions of multiple base models, using a meta-model to generate the final output, thereby enhancing predictive performance through the integration of diverse algorithmic approaches.

3.5 Training and evaluation of model

Each classifier, including the hard voting, soft voting, and stacking models, was trained on the training dataset. The trained models were then evaluated using the test dataset, and various performance metrics were calculated to assess their effectiveness. The evaluation process consisted of the following steps:

3.5.1 Training individual models

Each base model was independently trained on the training dataset to establish a baseline performance.

3.5.2 Training the ensemble models

The ensemble methods, including hard voting, soft voting, and stacking, were trained on the same training dataset. The performance of the ensemble models was computed by leveraging the outputs of the base models as estimators, thereby aggregating their predictive capabilities. Figure 4 shows the detailed architecture of the proposed methodology.

4 Result and discussion

In this study, the performance of the base model algorithms was evaluated using metrics such as the confusion matrix and accuracy score. The confusion matrix provides a comprehensive analysis of the model's performance by detailing the counts of true positives, false positives, true negatives, and false negatives for each attack category. This metric facilitates an in-depth understanding of the model's ability to differentiate between benign and anomalous traffic. Furthermore, accuracy is employed as an overall performance metric, reflecting the model's capability to classify both normal and malicious network activities correctly.

The results, illustrated in Fig. 5, demonstrate that the Random Forest model (CICIDS2017 dataset) on performs robustly overall, particularly in accurately identifying the BENIGN class, with 5686 instances correctly classified. However, 303 BENIGN samples are misclassified as PortScan, potentially due to similarities in the patterns of these two traffic types. The model also handles DoS Hulk attack traffic effectively, with 2550 instances accurately classified, although a small number of samples are misclassified as BENIGN. For the DDoS attack category, the model correctly predicts 1399 instances, with minor misclassifications into BENIGN or other attack types. Additionally, misclassifications are more pronounced for smaller attack categories such as Bot and DoS Slowhttptest, likely due to their limited representation in the dataset. This observation highlights the need for improved strategies to

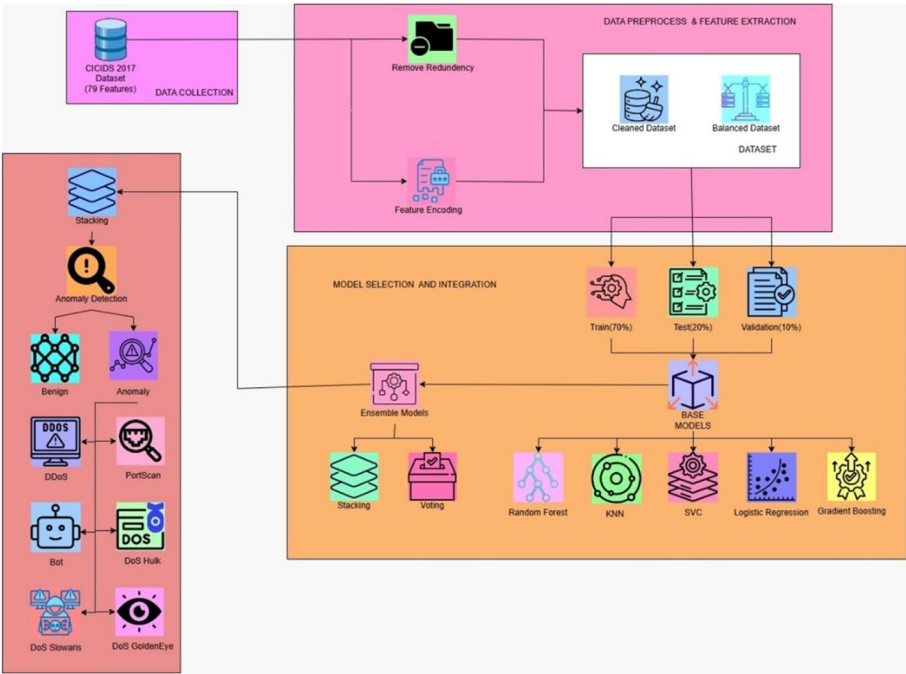


Fig. 4 Architecture for proposed anomaly detection using ML

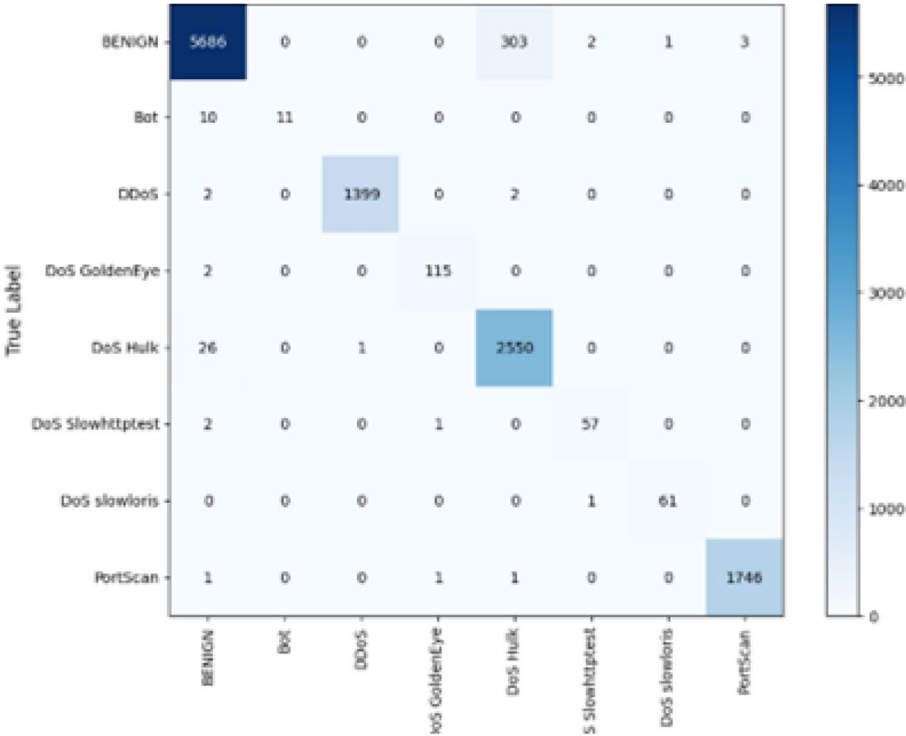


Fig. 5 Confusion matrix using random forest classifier (CICIDS2017 dataset)

handle underrepresented classes to enhance the model's overall performance and reliability. Figure 6, illustrates the Random Forest model on the other dataset (NFS-2023-TE).

Figure 7 presents the performance of the Support Vector Classifier (SVC) model for CICIDS2017 dataset, which demonstrates notable accuracy in classifying BENIGN traffic, with 5648 correct predictions, and

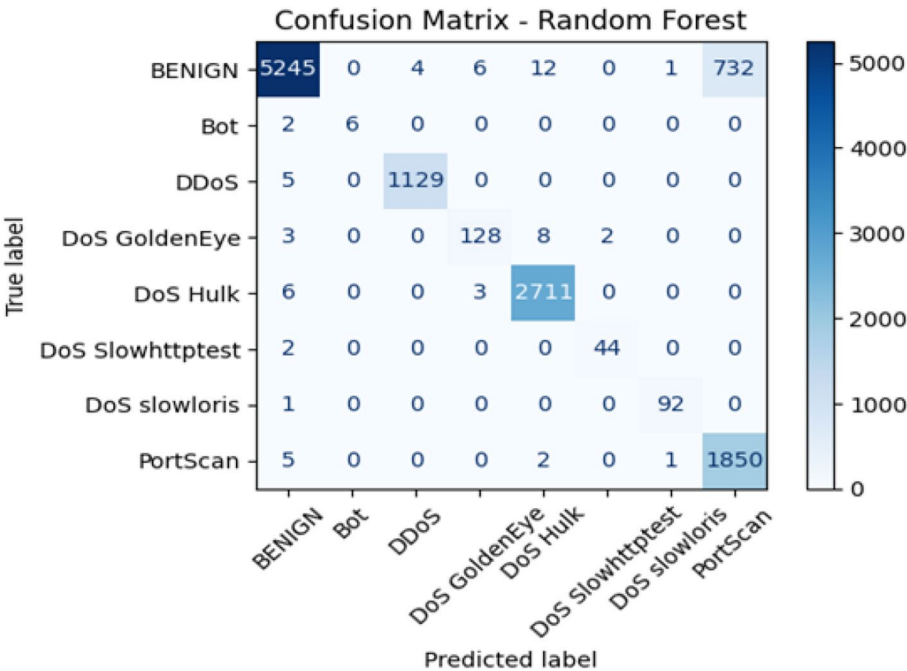


Fig. 6 Confusion matrix using random forest classifier (NFS-2023-TE dataset)

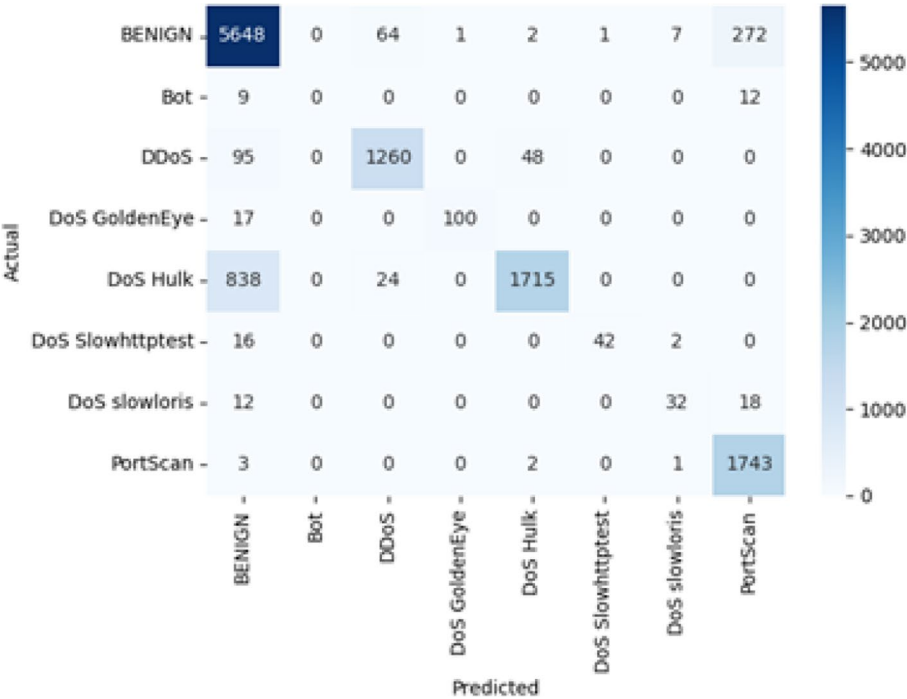


Fig. 7 Confusion matrix using support vector classifier (CICIDS2017 dataset)

PortScan, with 1743 accurate classifications. However, the model exhibits significant challenges in handling the DoS Hulk category, misclassifying 838 instances as BENIGN, which underscores its difficulty in distinguishing between these classes. For the DDoS category, the SVC achieves 1260 correct predictions but misclassifies 95 samples as BENIGN and 48 as other attack types, indicating a lack of precision in differentiating DDoS from traffic with similar patterns. Smaller attack categories, such as DoS Slowhttptest and DoS Slowloris, also pose difficulties, with instances frequently misclassified as either BENIGN or other attack types. These results suggest that while the SVC model performs effectively for larger, more distinct classes, it struggles to generalize across all attack types, particularly for underrepresented categories, highlighting a need for improved handling of minority classes. Similarly Fig. 8, illustrates the Support Vector Classifier model on the other dataset (NFS-2023-TE).

Figure 9 illustrates the performance of the logistic regression model on CICIDS2017 dataset, which exhibits a decline in accuracy compared to the K-nearest neighbors (KNN) model, particularly in distinguishing DDoS attacks. Out of 1403 instances, only 967 are correctly identified. The model also struggles with the DoS Hulk category, misclassifying 845 instances as BENIGN, which underscores its limitations in differentiating these traffic types. While the model performs reasonably well for PortScan and DoS Slowhttp, it demonstrates notable weaknesses in identifying other attack categories. This

indicates a need for enhanced sensitivity to diverse attack patterns to improve the model's overall performance and reliability. Similarly Fig. 10, illustrates the logistic regression model on the other dataset (NFS-2023-TE).

Figure 11 presents the performance of the gradient boosting model on the CICIDS2017 dataset, which demonstrates strong classification accuracy for the BENIGN and PortScan categories, with 5651 and 1746 correctly classified instances, respectively. However, 331 BENIGN samples are misclassified as PortScan, potentially due to overlapping features between these two traffic types. The model also achieves high accuracy for larger attack categories, such as DoS Hulk (2549 correct predictions) and DDoS (1397 correct predictions), with only minor misclassifications observed. In contrast, smaller attack classes, such as DoS GoldenEye and Bot, exhibit more significant classification errors. For example, only 6 out of 21 Bot instances are correctly identified. These findings indicate that while the gradient boosting model performs effectively for majority classes, it encounters difficulties with rare attack types, likely due to the imbalanced distribution of data in the dataset. Addressing this imbalance could improve the model's performance for underrepresented categories. Similarly Fig. 12, illustrates the Gradient Boosting model on the other dataset (NFS-2023-TE).

Figure 13 illustrates the performance of the K-nearest neighbors (KNN) model using the CICIDS2017 dataset, which demonstrates strong accuracy for BENIGN traffic, correctly classifying 5896 instances with minimal misclassifications, such as 82 BENIGN samples incorrectly

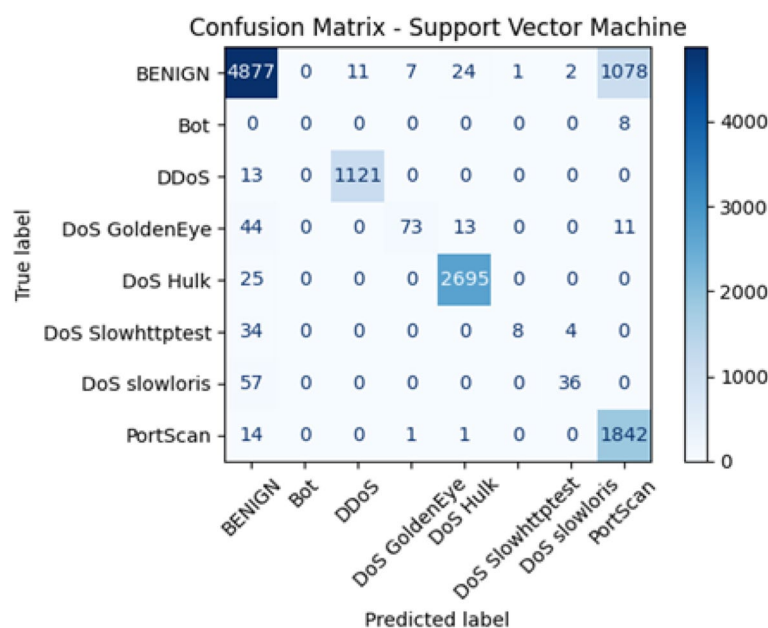


Fig. 8 Confusion matrix using support vector classifier (NFS-2023-TE dataset)

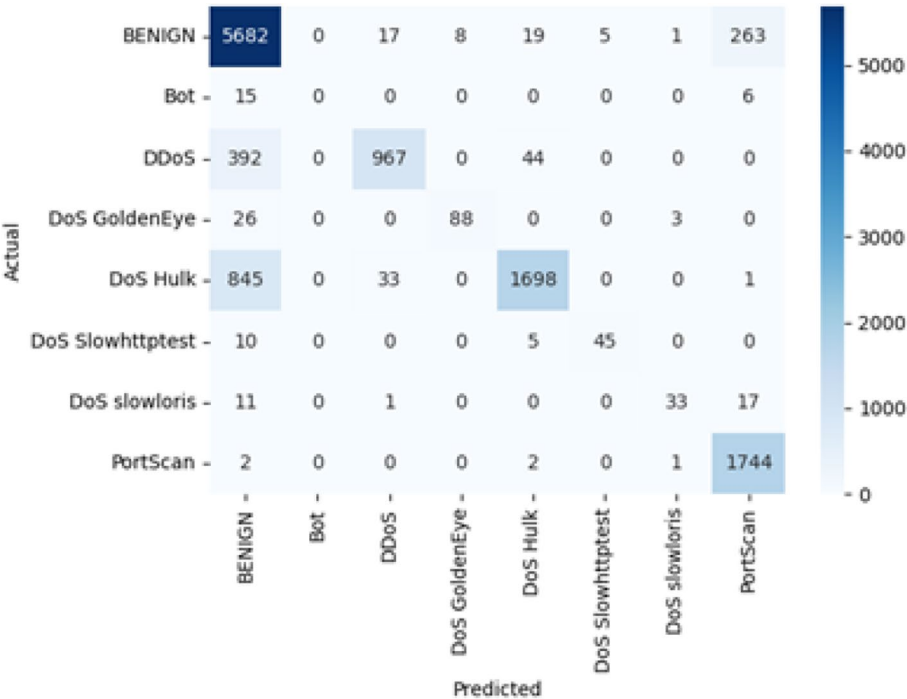


Fig. 9 Confusion matrix using logistic regression (CICIDS2017 dataset)

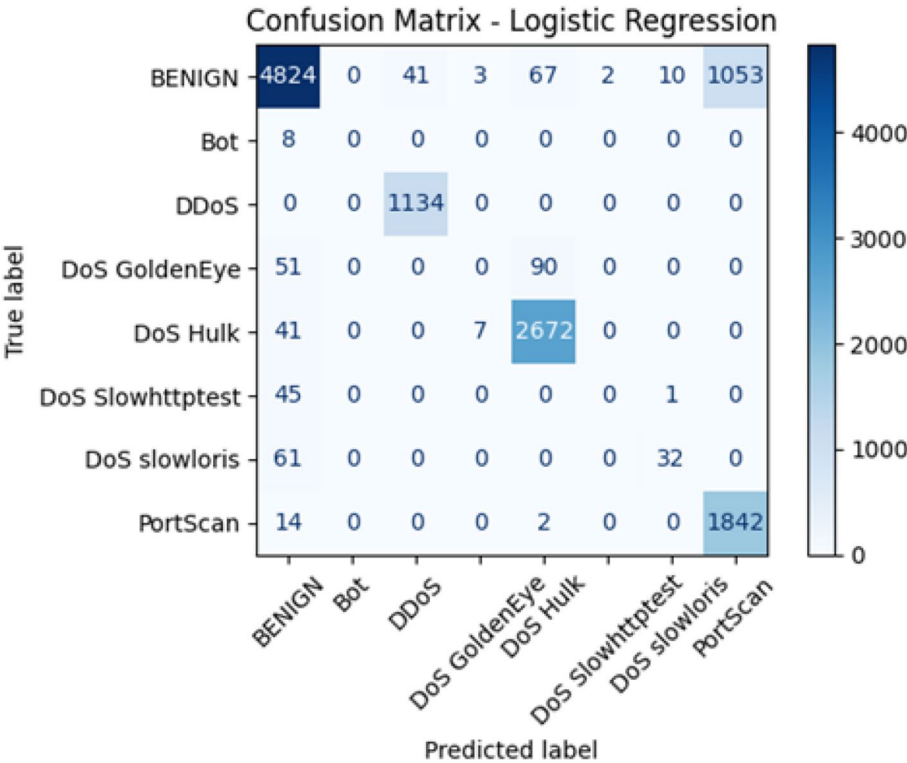


Fig. 10 Confusion matrix using logistic regression (NFS-2023-TE dataset)

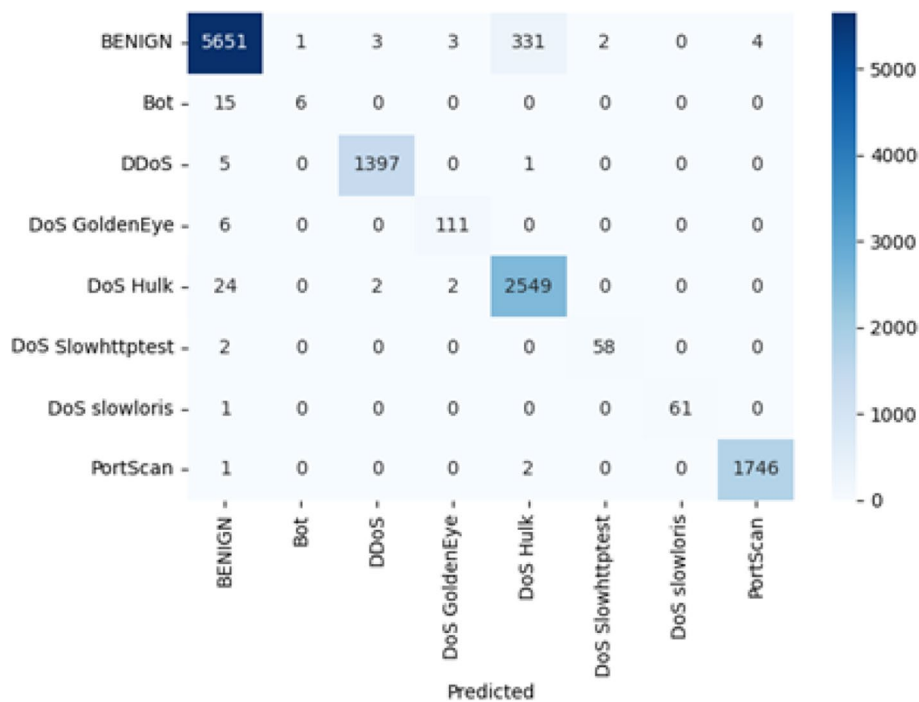


Fig. 11 Confusion matrix using gradient boosting (CICIDS2017 dataset)

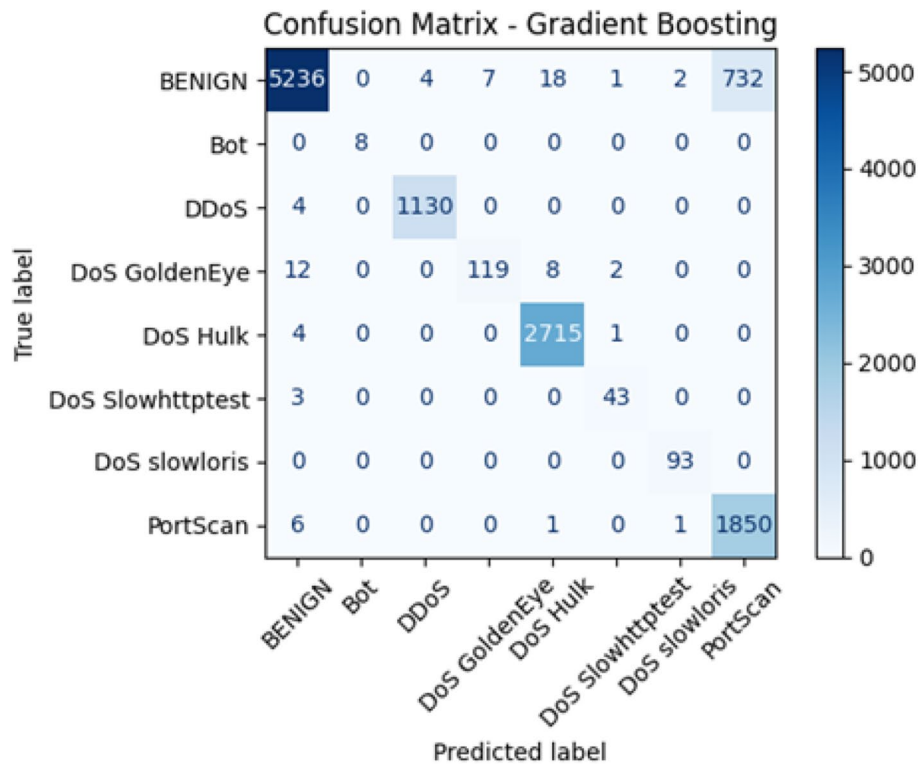


Fig. 12 Confusion matrix using gradient boosting (NFS-2023-TE dataset)

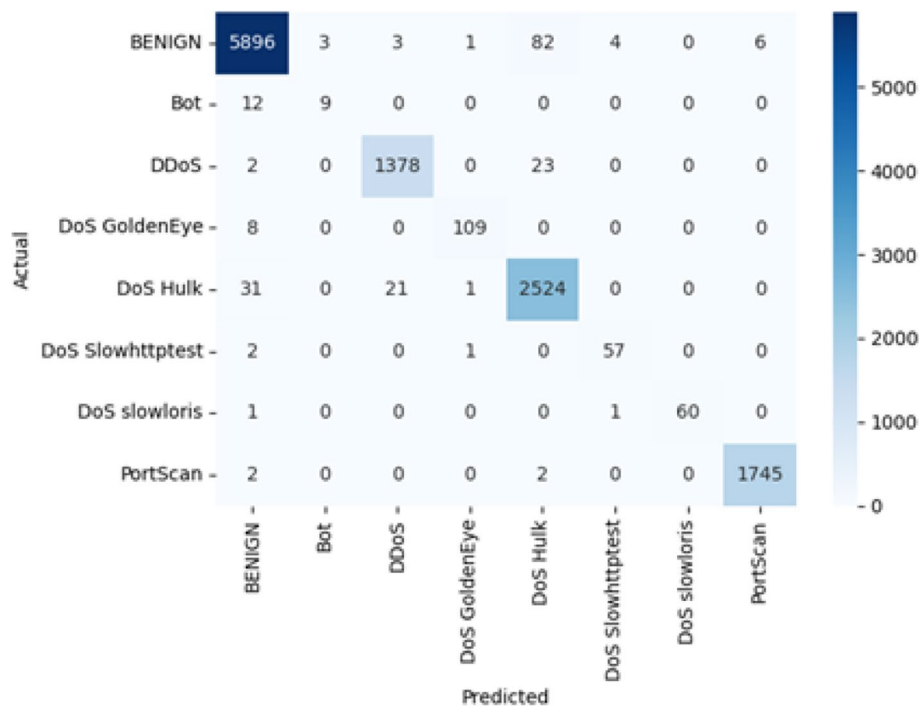


Fig. 13 Confusion matrix using K-nearest neighbors classifier (CICIDS2017 dataset)

identified as DoS Hulk. The model also performs well in detecting DDoS attacks, with 1378 instances accurately classified. However, the model exhibits notable misclassification for smaller attack categories. For example, only 57 and 60 instances of DoS Slowhttptest and DoS Slowloris are correctly classified, respectively. The Bot category poses a particular challenge, with only 9 instances accurately identified while 12 are misclassified. Overall, the KNN model demonstrates strong performance for larger and more distinct classes, such as BENIGN and DDoS, but faces significant difficulties in handling rare or similar attack types, highlighting the need for strategies to improve its classification of minority classes. Similarly Fig. 14, illustrates the KNN model on the other dataset (NFS-2023-TE).

Table 1 shows the comparison of the performance of different base models used on two different datasets (Dataset 1- CICIDS2017, Dataset 2 -NFS-2023-TE). The results indicate that the K-nearest neighbors (98.28%, 91%), random forest (97.00%, 96%), and gradient boosting (96.62%, 92%) models achieved high accuracy, demonstrating their effectiveness in network anomaly detection due to their capability to capture complex patterns. The support vector machine exhibited moderate performance, achieving an accuracy of 87.95%, likely due to challenges in handling overlapping classes. Logistic regression, with an accuracy of 85.59%, recorded the lowest performance among the evaluated models. To

improve overall model performance and accuracy, this study implements ensemble approaches utilizing voting and stacking methods. The voting approach includes both hard voting and soft voting, leveraging the strengths of individual models to enhance predictive capabilities.

Figure 15 highlights the performance of the hard voting classifier (CICIDS2017 dataset), which demonstrates strong classification accuracy across multiple categories, as evidenced by the high values along the diagonal of the confusion matrix. Notably, the model correctly classifies 5687 instances of the BENIGN class and 2549 instances of the DoS Hulk class. While misclassifications are present, they are relatively minimal. For instance, 302 BENIGN instances are misclassified as DDoS, with a few additional errors distributed across other classes. For attack types such as PortScan, the classifier achieves strong performance with 1746 correct predictions and only a small number of misclassifications. Overall, the model exhibits high accuracy, particularly for majority and well-defined classes. However, some confusion persists in distinguishing between overlapping or less frequent attack categories. Performance metrics for this classifier are also computed and presented in Table 2 to provide a comprehensive evaluation. Further, Fig 16 highlights the performance of the hard voting classifier for NFS-2023-TE dataset.

The performance metrics demonstrate high effectiveness across most attack types in the anomaly detection

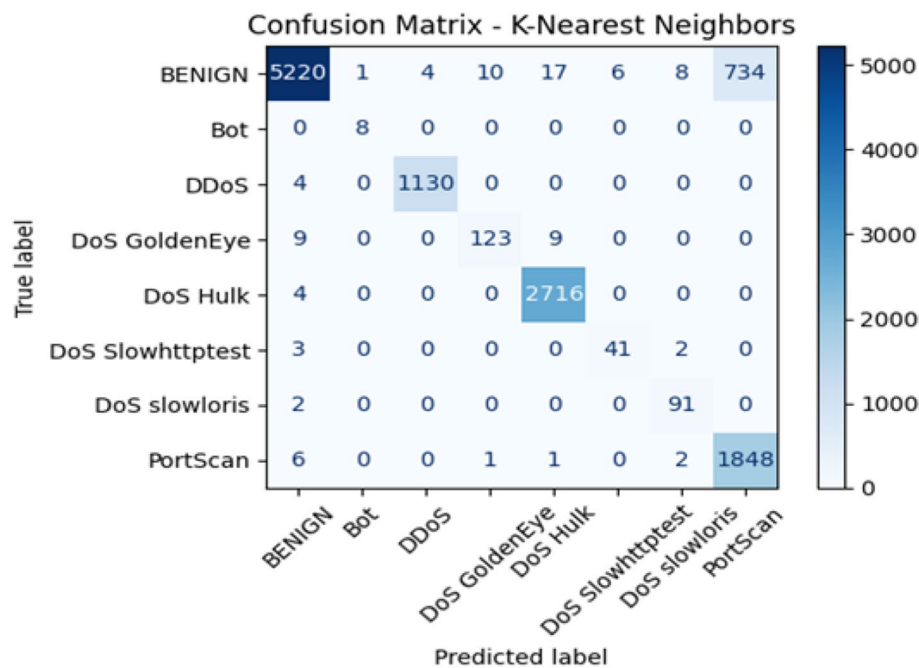


Fig. 14 Confusion matrix using K-nearest neighbors classifier (NFS-2023-TE dataset)

Table 1 Performance comparison of accuracy for base models

Model	Accuracy	
	Dataset 1	Dataset 2
Random forest	0.97	0.96
Support vector machine	0.88	0.90
K-nearest neighbors	0.98	0.91
Gradient boosting	0.97	0.92
Logistic regression	0.86	0.93

system as presented in Table 2 for both datasets (DS1 - CICIDS2017, DS2 - NFS-2023-TE). The comparison of the F1-Score and precision of hard Voting model is shown in Fig. 17 for DS1. BENIGN traffic is classified with strong precision (0.99), recall (0.95), and F1-score (0.97), indicating robust detection. Both DDoS and PortScan categories achieved perfect scores (1.00), signifying flawless detection for these attack types. DoS GoldenEye, DoS Hulk, and DoS Slowloris also exhibited strong performance, with F1-scores of 0.97, 0.94, and 0.99, respectively, reflecting consistent accuracy across these categories. Similarly, Fig. 18 demonstrates the comparison of F1 Score and precision for another dataset (NFS-2023-TE dataset).

DoS Slowhttptest achieved a solid F1-score of 0.95, further underscoring the model's overall reliability. However, the model faces challenges in detecting Bot traffic, which shows a low recall of 0.48 and an F1-score of 0.65.

Despite achieving perfect precision (1.00) for Bot traffic, the model struggles to identify all instances of this attack type. Overall, the model performs excellently across the majority of categories, with minor challenges in detecting Bot traffic, indicating areas for further improvement in handling rare or less distinct attack types.

Figure 19 illustrates the performance of the soft voting classifier using CICIDS2017 dataset, which demonstrates strong accuracy across multiple classes, with high values along the diagonal. For example, the model correctly classifies 5,906 instances of the BENIGN class and 2550 instances of the DoS Hulk class. Misclassifications are slightly reduced compared to the hard voting classifier, with only 84 BENIGN instances misclassified as DDoS, suggesting an improved ability to distinguish between these two classes. Further, Fig. 20 illustrates the performance of the soft voting classifier using NFS-2023-TE dataset.

Attack categories such as DDoS and PortScan also exhibit high accuracy, with 1398 and 1746 correct predictions, respectively, and fewer misclassifications. Overall, the soft voting classifier leverages probability-weighted decisions, resulting in a slightly better performance in handling ambiguous or overlapping samples. This enhances its robustness across all classes, improving classification accuracy, particularly in cases where the attack types exhibit similarities.

The performance metrics demonstrate excellent effectiveness across most attack types in the anomaly

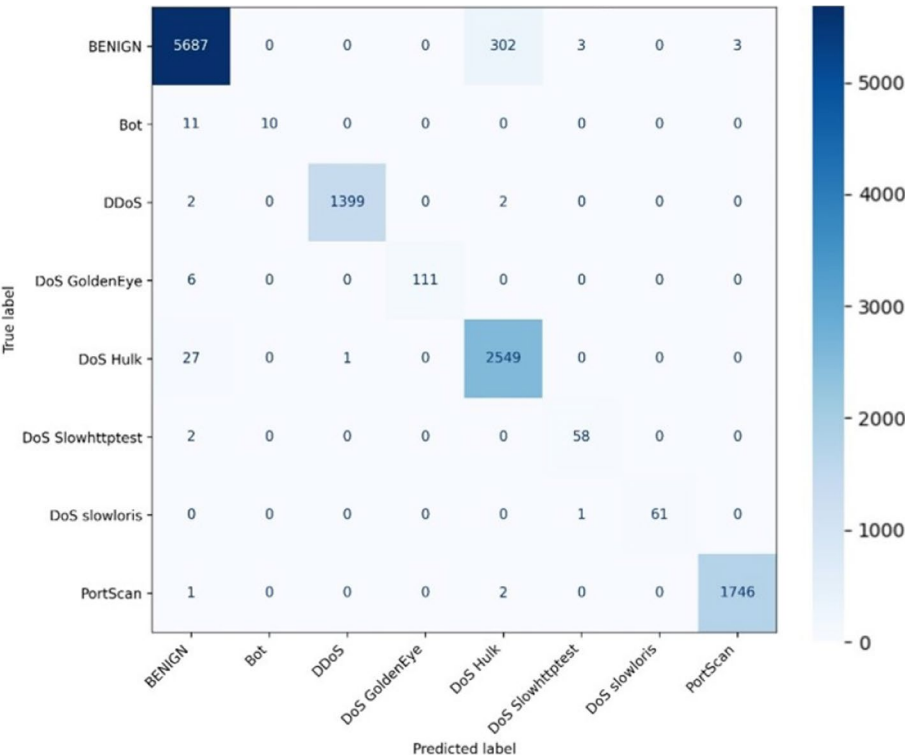


Fig. 15 Hard voting confusion matrix (CICIDS2017 dataset)

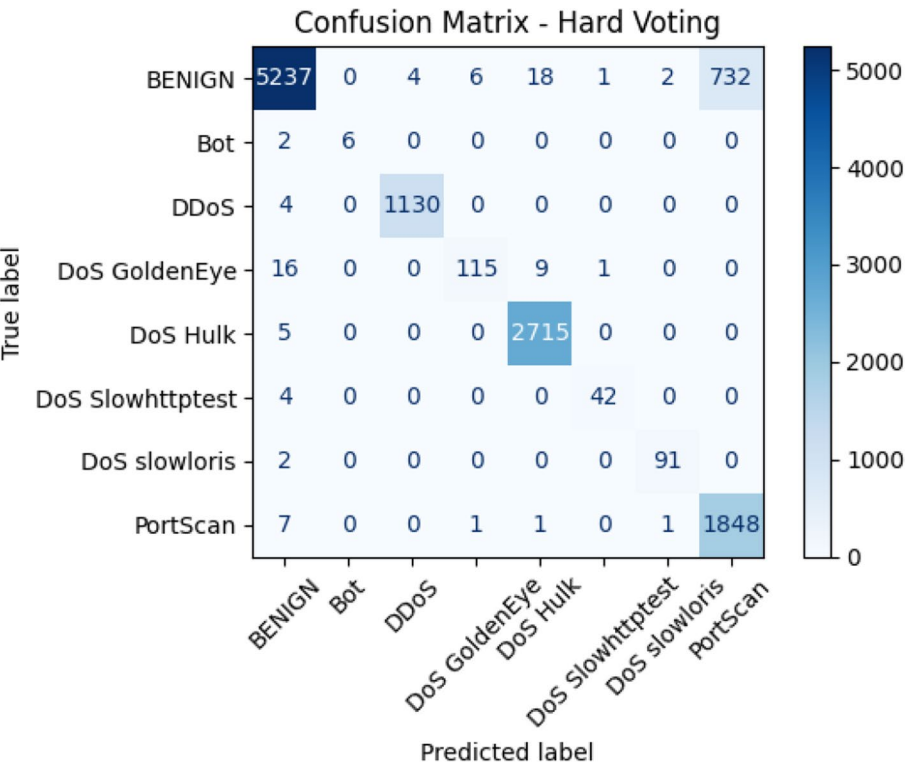


Fig. 16 Hard voting confusion matrix (NFS-2023-TE dataset)

Table 2 Hard voting classification report

Attacks	Precision		Recall		F1-score		Support	
	DS1	DS2	DS1	DS2	DS1	DS2	DS1	DS2
BENIGN	0.99	0.99	0.95	0.87	0.97	0.93	5995	6000
Bot	1	1	0.48	0.75	0.65	0.86	21	8
DDoS	1	1	1	1	1	1	1403	1134
DoS GoldenEye	1	0.94	0.95	0.82	0.97	0.87	117	141
DoS Hulk	0.89	0.99	0.99	1	0.94	0.99	2577	2720
DoS Slowhttptest	0.94	0.95	0.97	0.91	0.95	0.93	60	46
DoS slowloris	1	0.97	0.98	0.98	0.99	0.97	62	93
PortScan	1	0.72	1	0.99	1	0.83	1749	1858

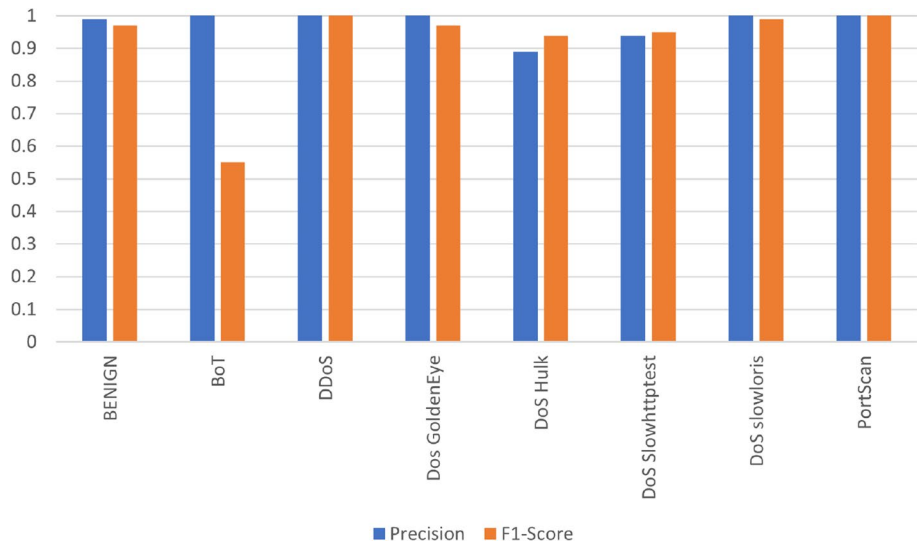


Fig. 17 F1-score and precision of hard voting model (CICIDS2017 dataset)

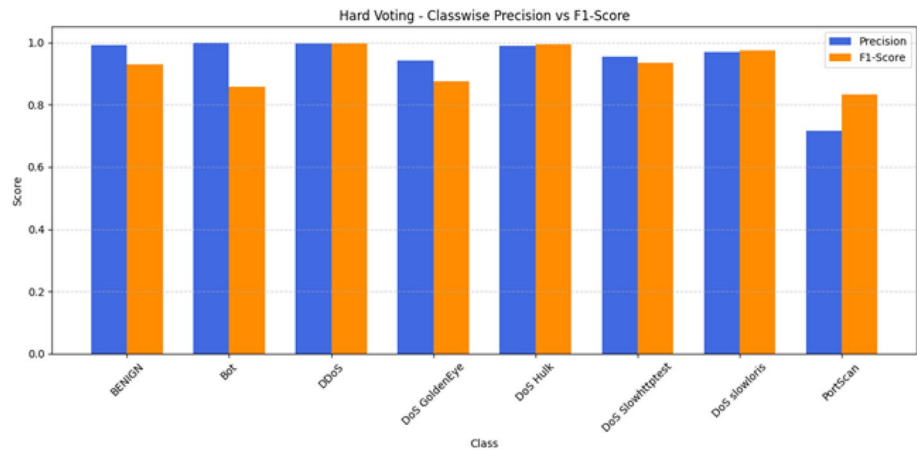


Fig. 18 F1-score and precision of hard voting model (NFS-2023-TE dataset)

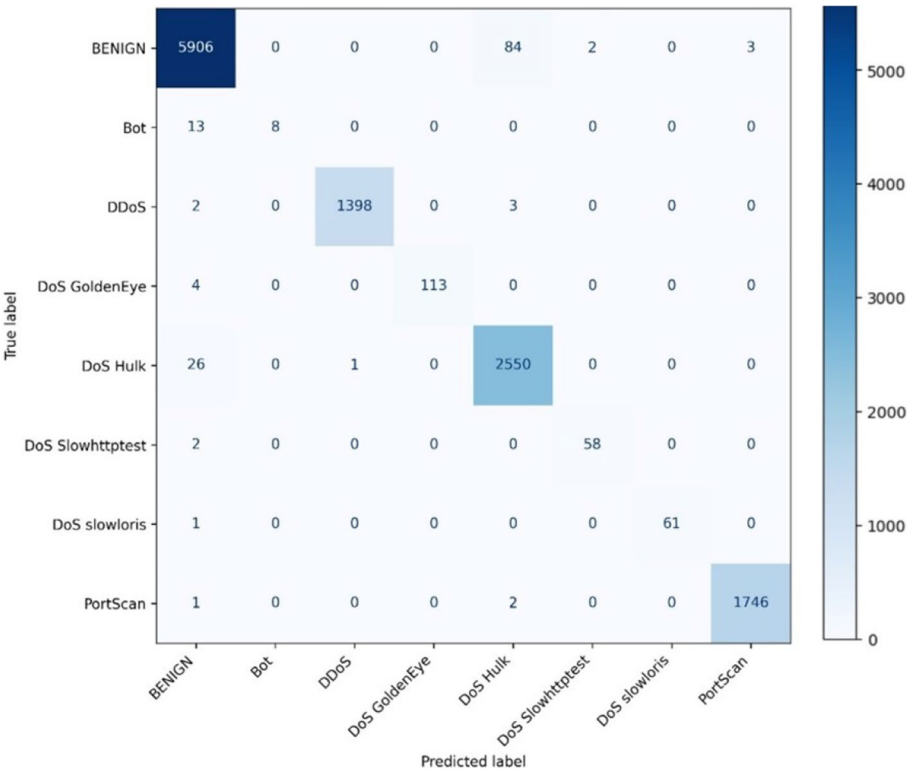


Fig. 19 Soft voting confusion matrix (CICIDS2017 dataset)

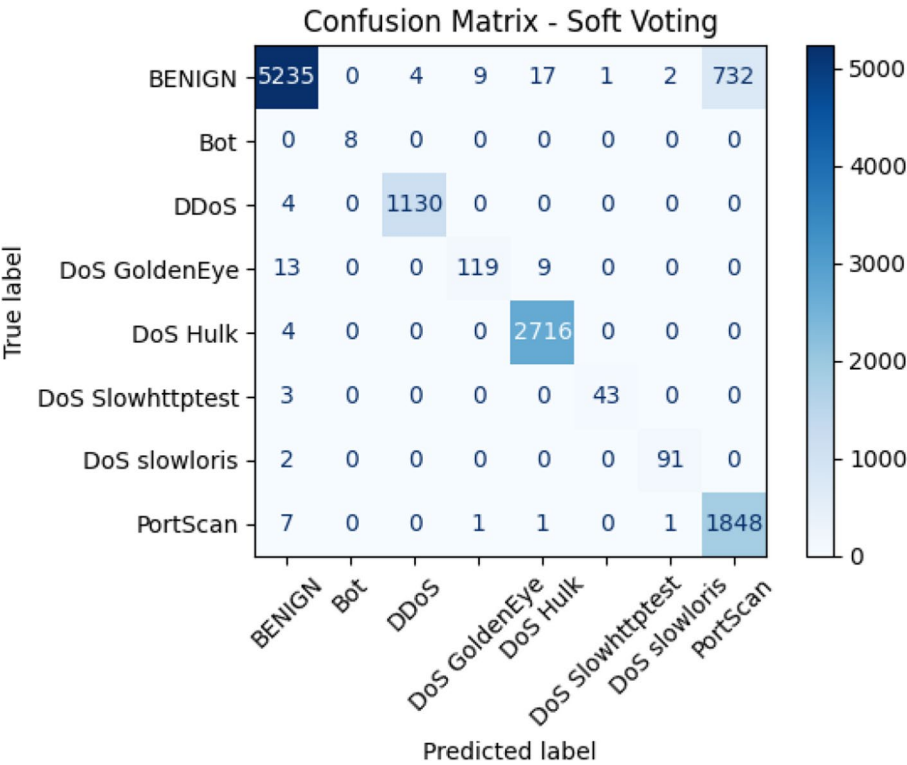


Fig. 20 Soft voting confusion matrix (NFS-2023-TE dataset)

detection system for both datasets (DS1 - CICIDS2017, DS2 - NFS-2023-TE) as presented in Table 3. The comparison of the F1-score and precision of the soft Voting model is shown in Fig. 21 for the CICIDS2017 dataset. BENIGN traffic is nearly perfectly classified, with precision, recall, and F1-score values of 0.99, indicating robust detection. Both DDoS and PortScan categories achieved perfect scores (1.00), reflecting flawless detection. Similarly, DoS GoldenEye, DoS Hulk, and DoS Slowloris exhibited high accuracy, with F1-scores of 0.98, 0.98, and 0.99, respectively, signifying reliable detection for these attack types. Similarly, Fig. 22 demonstrates the

comparison of F1 Score and precision for another dataset (NFS-2023-TE dataset). DoS Slowhttptest also performed well, with a balanced precision, recall, and F1-score of 0.97. However, the model faces challenges with Bot traffic, which shows a low recall (0.38) and F1-score (0.55), despite achieving perfect precision (1.00). This indicates difficulty in identifying all instances of Bot attacks. Overall, the system demonstrates robustness across most categories, with minor difficulties in handling Bot traffic, suggesting areas for improvement in detecting rare or less distinct attack types.

Table 3 Soft voting classification report

Attacks	Precision		Recall		F1-score		Support	
	DS1	DS2	DS1	DS2	DS1	DS2	DS1	DS2
BENIGN	0.99	0.99	0.99	0.87	0.99	0.93	5995	6000
Bot	1	1	0.38	1	0.55	1	21	8
DDoS	1	1	1	1	1	1	1403	1134
DoS GoldenEye	1	0.92	0.97	0.84	0.98	0.88	117	141
DoS Hulk	0.97	0.99	0.98	1	0.98	0.99	2577	2720
DoS Slowhttptest	0.97	0.98	0.97	0.93	0.97	0.96	60	46
DoS slowloris	1	0.97	0.98	0.98	0.99	0.97	62	93
PortScan	1	0.72	1	0.99	1	0.83	1749	1858

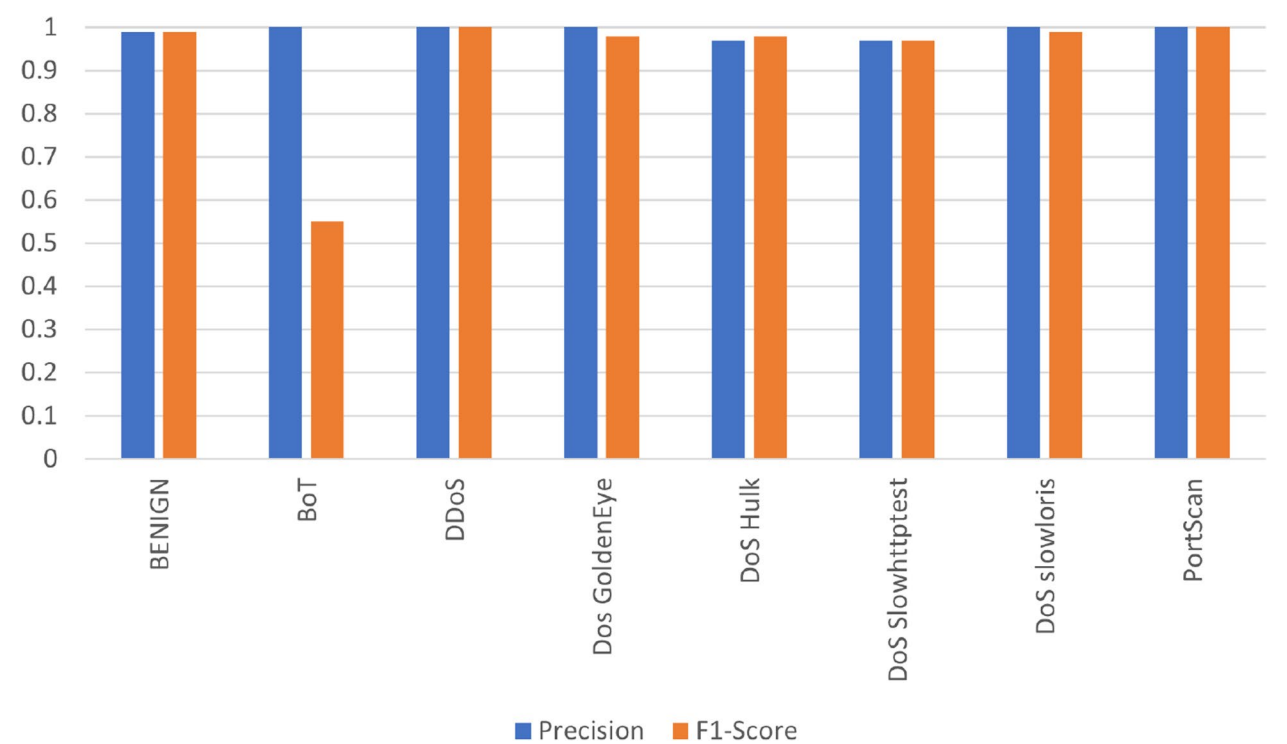


Fig. 21 F1-score and precision of soft voting model (CICIDS2017 dataset)

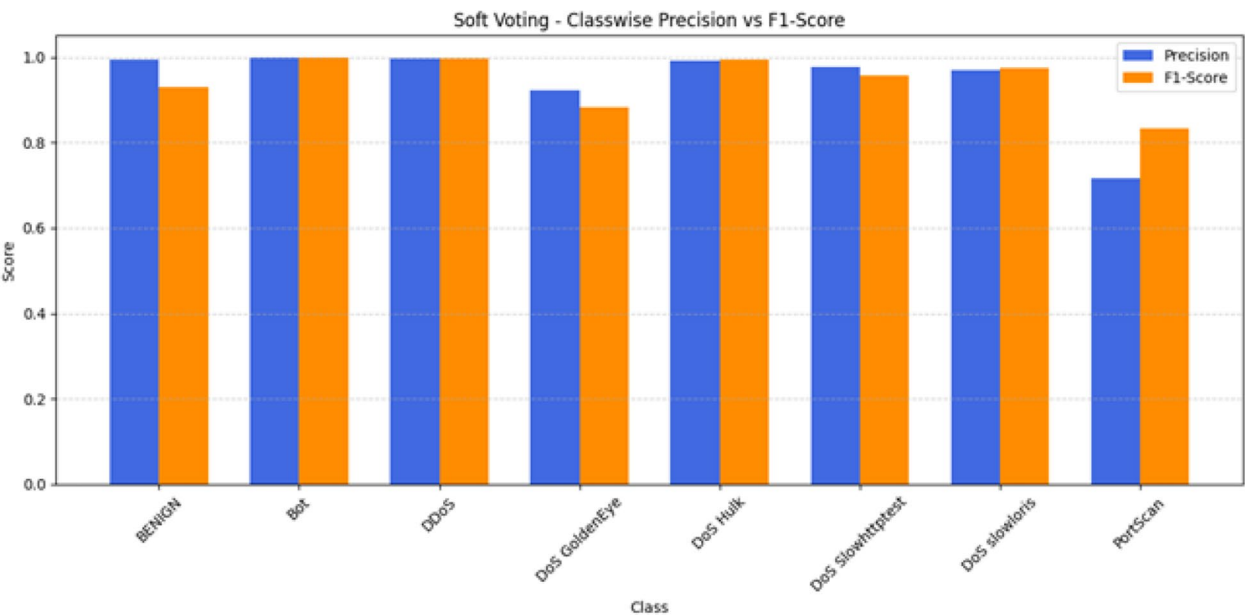


Fig. 22 F1-score and precision of soft voting model (NFS-2023-TE dataset)

Figure 23 illustrates that the stacking model performs effectively, achieving high accuracy for the BENIGN (5918 correct predictions), DoS Hulk (2527 correct predictions), and PortScan (1746 correct predictions) categories. However, some confusion is observed between Class 3 and Class 4, indicating challenges in distinguishing these closely related classes. Overall, this ensemble approach delivers balanced performance,

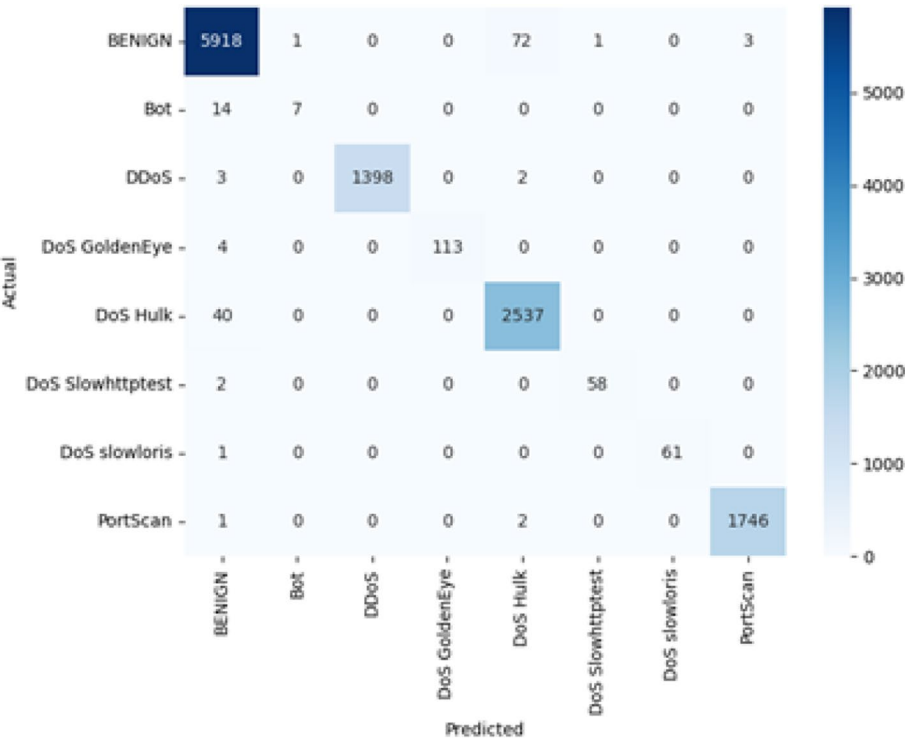


Fig. 23 Stacking confusion matrix (CICIDS2017 dataset)

effectively capturing primary patterns. Further optimization could enhance its ability to differentiate between these similar classes, improving classification precision. Figure 24 demonstrates stacking model performance on other dataset.

The performance metrics demonstrate excellent accuracy across most attack types in the anomaly detection system, as presented in Table 4. The comparison of the F1-score and precision of the stacking model for the CICIDS2017 dataset is shown in Fig. 25. BENIGN traffic is classified nearly perfectly, with precision, recall,

and F1-score values of 0.99, indicating robust detection. Both DDoS and PortScan achieved flawless detection, attaining perfect scores (1.00) across all metrics. Similarly, DoS GoldenEye, DoS Hulk, and DoS Slowloris exhibited strong performance, with F1-scores of 0.98, 0.98, and 0.99, respectively, reflecting the model's consistent reliability for these attack types. Similarly, Fig. 26 demonstrates the comparison of F1 Score and precision of the stacking model for another dataset (NFS-2023-TE dataset)

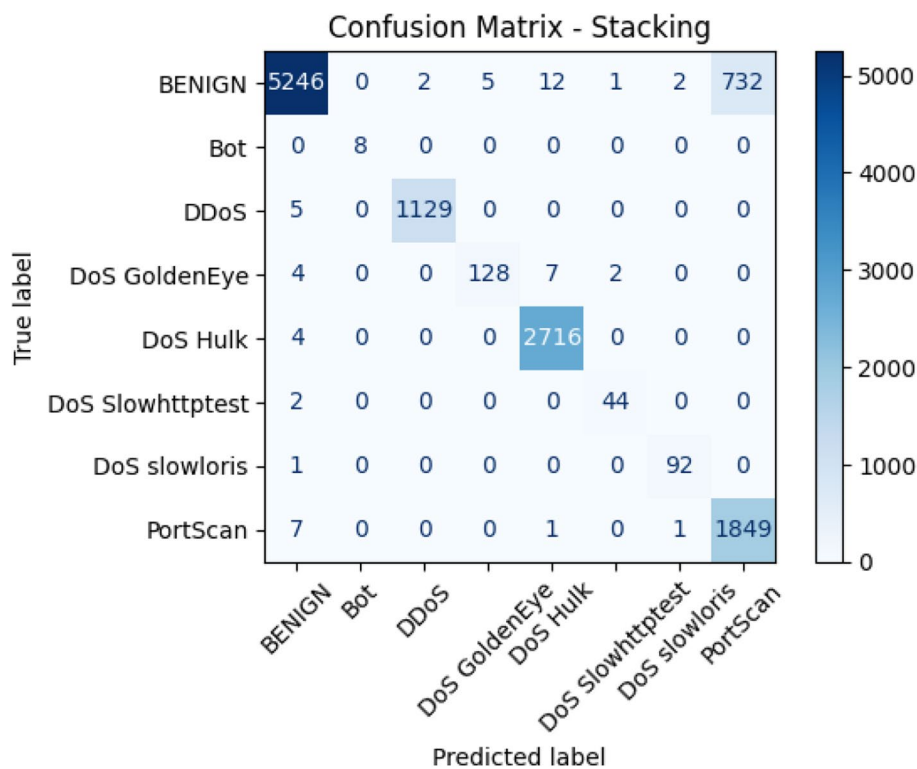


Fig. 24 Stacking confusion matrix (NFS-2023-TE dataset)

Table 4 Stacking classification report

Attacks	Precision		Recall		F1-score		Support	
	DS1	DS2	DS1	DS2	DS1	DS2	DS1	DS2
BENIGN	0.99	1	0.99	0.87	0.99	0.93	5995	6000
Bot	1	1	0.38	1	0.55	1	21	8
DDoS	1	1	1	1	1	1	1403	1134
DoS GoldenEye	1	0.96	0.97	0.91	0.98	0.93	117	141
DoS Hulk	0.97	0.99	0.98	1	0.98	1	2577	2720
DoS Slowhttptest	0.97	0.94	0.97	0.96	0.97	0.95	60	46
DoS slowloris	1	0.97	0.98	0.99	0.99	0.98	62	93
PortScan	1	0.72	1	1	1	0.83	1749	1858

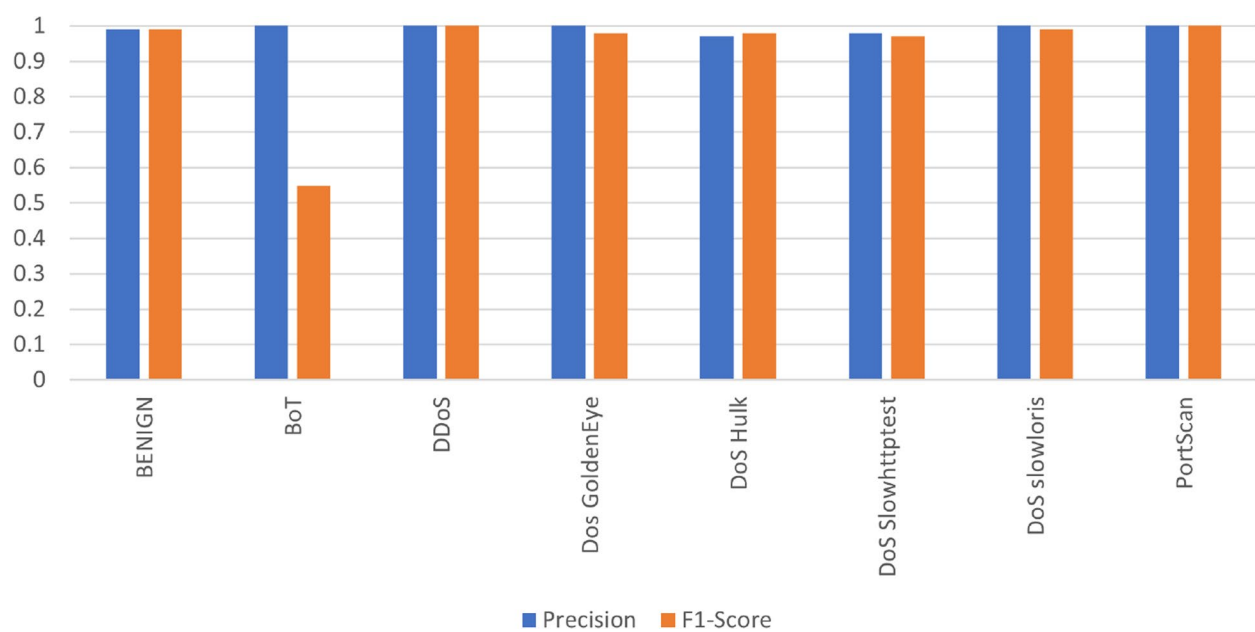


Fig. 25 F1-score and precision of stacking model (CICIDS2017 dataset)

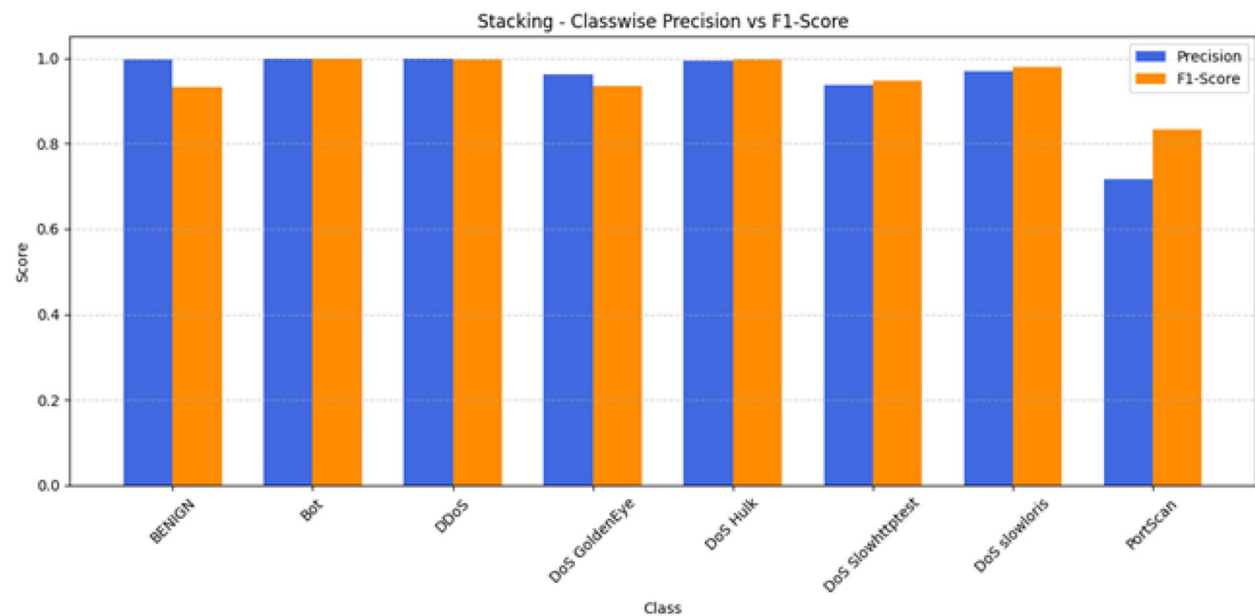


Fig. 26 F1-score and precision of stacking model (NFS-2023-TE dataset)

DoS Slowhttptest also performed well, with an F1-score of 0.97, supported by high precision (0.98) and recall (0.95). However, the model faces challenges with Bot traffic, as evidenced by its low recall (0.38) and F1-score (0.55), despite achieving perfect precision (1.00). This suggests difficulty in detecting all instances of this attack type. Overall, the system demonstrates robustness and

high effectiveness across most categories, with minor limitations in handling Bot traffic, indicating potential areas for further improvement in detecting rare or similar attack types. Figure 27 demonstrates the accuracy scores of various classifiers and ensemble methods employed in the classification task for CICIDS2017 dataset. Individual

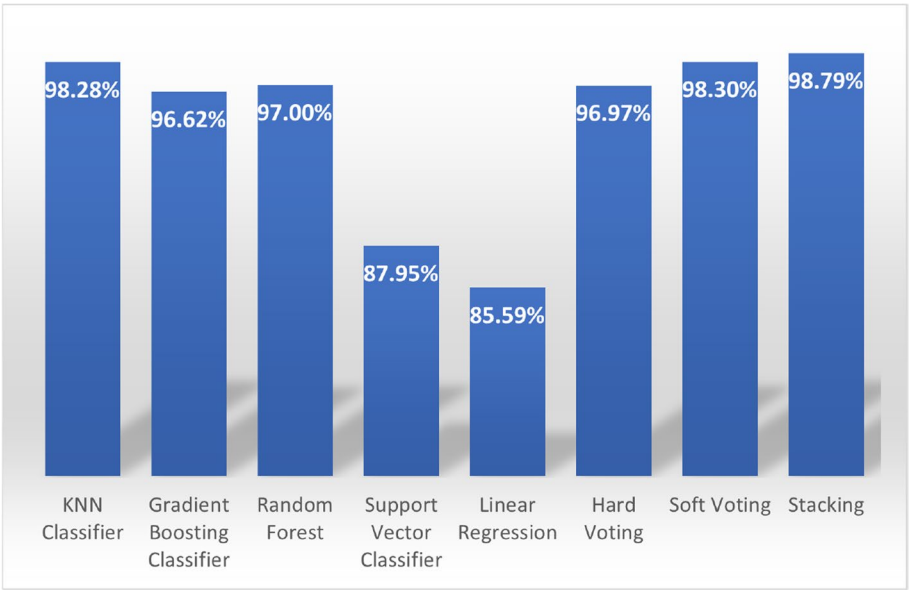


Fig. 27 Comparison of accuracy for all the methods used for CICIDS2017 dataset

classifiers, including K-nearest neighbors (KNN), gradient boosting, and random forest, achieved high accuracy, ranging from 96 to 98%. The ensemble methods, specifically soft voting and stacking, demonstrated slightly improved performance, with stacking achieving the highest accuracy at 98.79%. Further, Fig. 28 demonstrates the accuracy scores of various classifiers and ensemble methods employed in the classification task for NFS-2023-TE dataset.

5 Conclusion and future work

This research highlights the increasing demand for advanced solutions in network security as cyberattacks continue to evolve in sophistication and complexity. The investigation into ensemble learning methods, particularly voting and stacking techniques, along with the introduction of a novel hybrid model, emphasizes the potential of combining multiple algorithms to achieve superior detection accuracy. By utilizing the diverse and

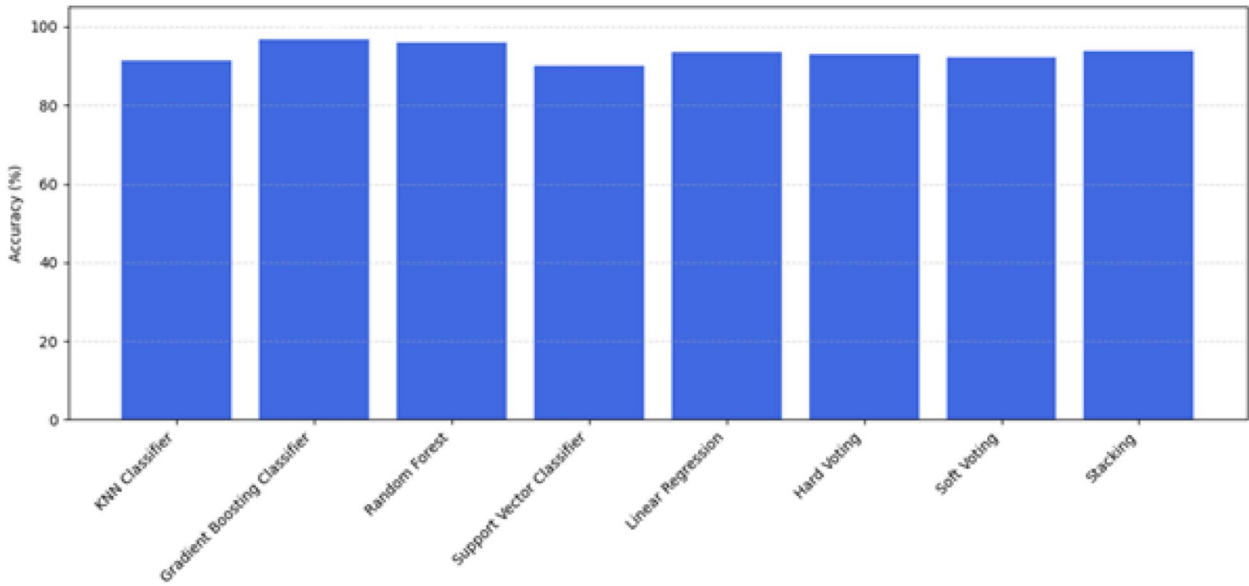


Fig. 28 Comparison of accuracy for all the methods used for NFS-2023-TE dataset

real-world CICIDS 2017 and NFS-2023-TE datasets, this study ensures that the models are robust and capable of addressing both historical and contemporary attack vectors. The thorough evaluation of model performance using standard metrics further reinforces the practical applicability of the research.

In summary, this study provides a solid foundation for future exploration in areas such as deep learning and real-time detection. As the landscape of network security continues to evolve, the findings of this research contribute to the development of intelligent network detection systems that are better equipped to defend against both current and emerging cyber threats.

Abbreviations

NDS	Network detection system
KNN	K-nearest neighbors
SVM	Support vector machines
GB	Gradient boosting
RF	Random forest
LR	Logistic regression
DoS	Denial of service
DDoS	Distributed Denial of Service
APT	Advanced Persistent Threat
HTTP	Hypertext Transfer Protocol
FL	Federated learning

Authors' contributions

All authors have made significant and direct intellectual contributions to this work. The contributions are as follows: VP: conceptualization, investigation, methodology, data collection, formal analysis, writing—original draft preparation, and project administration. KP: supervise, validate, write, review, and edit. MS: methodology, data analysis, writing—review and editing, and visualization. VB and SH: methodology, experimental evaluation, and review of the manuscript. All authors have reviewed and approved the final version of the manuscript for publication.

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Data availability

No datasets were generated or analyzed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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